The Geography of R&D and the Onshoring of Innovation: Evidence from TCJA Section 174

Donald E. Bowen III Joan Farre-Mensa
Luke Stein

Philly Five
October 30, 2026



LEHIGH
UNIVERSITY

Section 174 and the Tax Treatment of R&D

- TCJA (2017) was primarily deficit financed
- A pay-for: §174 in TCJA replaces expensing of R&D with amortization
- §174 set to go into effect for tax years beginning after **Dec. 31, 2021**
  - Business community expected repeal during each Congress 2018–2021 (Cowx et al. 2025)
- Extra §174 goal: onshoring innovative activity by making domestic R&D cheaper than foreign R&D
  - **Domestic R&D:** 5-year amortization $\implies$ after-tax cost of R&D $\uparrow\approx 5.4\%$
  - **Foreign R&D:** 15-year amortization $\implies$ after-tax cost of R&D $\uparrow\approx 12.4\%$

Section 174 and the Tax Treatment of R&D

- TCJA (2017) was primarily deficit financed
- A pay-for: §174 in TCJA replaces expensing of R&D with amortization
- §174 set to go into effect for tax years beginning after **Dec. 31, 2021**
  - Business community expected repeal during each Congress 2018–2021 (Cowx et al. 2025)
- Extra §174 goal: onshoring innovative activity by making domestic R&D cheaper than foreign R&D
  - **Domestic R&D:** 5-year amortization $\implies$ after-tax cost of R&D $\uparrow\approx 5.4\%$
  - **Foreign R&D:** 15-year amortization $\implies$ after-tax cost of R&D $\uparrow\approx 12.4\%$

Research Question

How did the **quantity, location, and characteristics** of innovation outputs at U.S. firms respond to an exogenous, firm-specific shock to the **user cost of R&D**?

Empirical Design

$$Y_{igt} = \alpha_i + \gamma_{jt} + \beta\,(T_i \times Post_t) + \theta' W_{i,t-1} + \varepsilon_{igt}$$

- **Treatment T_i:** above-median share of foreign-located inventors on pre-TCJA patents (2010–2016)
  - Validation: 77% US-inventor share vs. 81% domestic R&D share in Census data (Glaeser et al. 2025)
- **Outcome Y_{igt}:** patent applications split by inventor location, $g \in \{$Foreign-only, US-only, Collaborative$\}$
- **Applications, not grants:** May FYE 2024 applications published Nov '25
- **Estimation:** Poisson; firm FE, industry $\times$ year FE, lagged firm controls

First Stage: R&D Falls Where the Foreign Penalty Bites

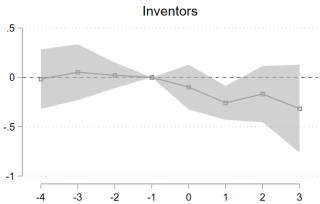
	R&D		CAPX		Inventors	
	(1)	(2)	(3)	(4)	(5)	(6)
$T_i \times Post_t$	-0.013**	-0.013**	0.001	0.001	-0.169**	-0.139**
	(0.006)	(0.006)	(0.001)	(0.001)	(0.072)	(0.071)
Log(Assets)		-0.006		0.001		0.250**
		(0.008)		(0.001)		(0.122)
MB		-0.003		0.001*		-0.034*
		(0.002)		(0.000)		(0.018)
Cash/Assets		0.009		0.011**		-0.714**
		(0.040)		(0.005)		(0.317)
Fixed effects:						
Firm	Yes	Yes	Yes	Yes	Yes	Yes
Ind x Year	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	4,623	4,618	4,623	4,618	3,538	3,534

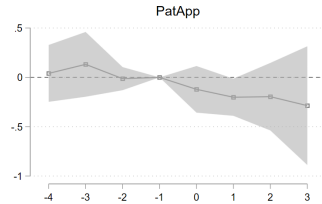
- R&D/assets falls by ≈1.3pp at treated firms (≈13% of the sample mean of 9.8%)
- Placebo: no differential change in CAPX ⇒ not a general investment contraction
- Total inventor headcount down ≈13–16% (Poisson)
- → the shock reaches real inventive activity; decompose by inventor geography next

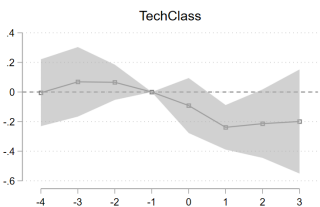
The Innovative Activities of Foreign Inventors Fell

	Inventors		PatApp		TechClasses	
	(1)	(2)	(3)	(4)	(5)	(6)
$T_i \times Post_t$	-0.245***	-0.202**	-0.187*	-0.103	-0.194**	-0.153*
	(0.094)	(0.098)	(0.111)	(0.120)	(0.077)	(0.078)
Log(Assets)		0.364**		0.809***		0.417**
		(0.174)		(0.251)		(0.172)
MB		-0.081***		-0.092**		-0.070**
		(0.026)		(0.037)		(0.029)
Cash/Assets		-0.697		-1.233		-0.540*
		(0.488)		(1.110)		(0.282)
Fixed effects:						
Firm	Yes	Yes	Yes	Yes	Yes	Yes
Ind x Year	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	2,740	2,739	2,750	2,749	2,252	2,252

The Innovative Activities of Foreign Inventors Fell







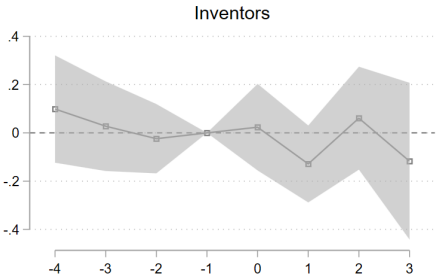
- 2017–2018 added to figures to show pre-trends
- Event time 1 = FY2022; −1 (FY2020) omitted
- Pre-trends flat; sharp break after 2022

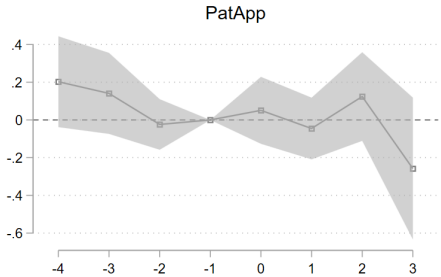
Domestic Innovative Activity Did Not Offset the Decline

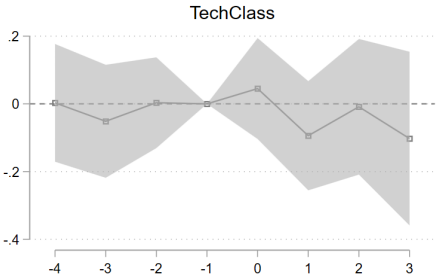
	Inventors		PatApp		TechClasses	
	(1)	(2)	(3)	(4)	(5)	(6)
$T_i \times Post_t$	-0.073	-0.052	-0.055	-0.029	-0.066	-0.050
	(0.073)	(0.070)	(0.082)	(0.080)	(0.064)	(0.065)
Log(Assets)		0.143		0.172*		0.119
		(0.101)		(0.100)		(0.081)
MB		0.000		0.004		0.014
		(0.014)		(0.014)		(0.014)
Cash/Assets		-0.665**		-0.718**		-0.321
		(0.300)		(0.301)		(0.239)
Fixed effects:						
Firm	Yes	Yes	Yes	Yes	Yes	Yes
Ind x Year	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	3,277	3,273	3,277	3,273	3,143	3,139

Upper 95% CI: rules out a domestic rise of more than $\sim$7–9% in inventors (vs. -18 to -22% for international)

Domestic Innovative Activity Did Not Offset the Decline







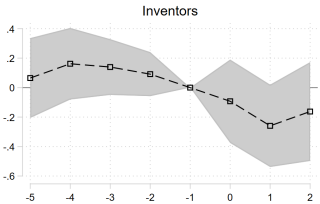
Within-Firm Triple Difference: International vs. Domestic

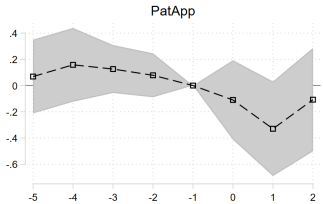
	Inventors		PatApp		TechClasses	
	(1)	(2)	(3)	(4)	(5)	(6)
$T_i \times Post_t \times Intl_g$	-0.202**	-0.239**	-0.194*	-0.254*	-0.144	-0.182**
	(0.101)	(0.113)	(0.117)	(0.138)	(0.088)	(0.092)
$T_i \times Post_t$	-0.045		-0.020		-0.059	
	(0.072)		(0.083)		(0.060)	
Fixed effects:						
Firm x Group	Yes	Yes	Yes	Yes	Yes	Yes
Ind x Year x Group	Yes	Yes	Yes	Yes	Yes	Yes
Firm x Year	No	Yes	No	Yes	No	Yes
Obs.	6,055	4,264	6,065	4,282	5,421	3,364

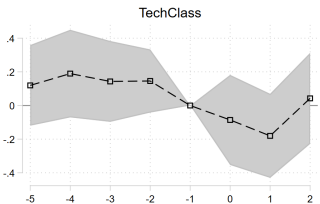
Firm×year FE absorb all firm-level shocks;
Stacked international and domestic outcomes;

$T_i \times Post_t \times Intl_g$ = extra effect on international activity.

Within-Firm Triple Difference: International vs. Domestic







- Coefficients on $T_i \times Intl_g \times$ event year
- Year-1 dip on all three outcomes, partly reverting in year 2
- Pre-period points are positive but jointly insignificant ($p = 0.57$ inventors, 0.69 applications, 0.53 technology classes)

Robustness

- **Estimator and functional form:** quantile (median), OLS $\log(1+y)$, OLS IHS
  - International effects stay negative; domestic stays small
- **China decoupling:** drop every patent with a China-located inventor
  - International effects essentially unchanged (inventors -0.251 vs. -0.245)
- **CHIPS Act:** subsidizes domestic inventors at treated firms
  - Works against our domestic null, so the null is conservative

Which Margins Matter: Productivity, Relocation, Attrition?

- **Productivity:** applications per foreign inventor $\uparrow \approx$ 36–38%; per domestic inventor flat
  - Consistent with culling lower-output foreign projects
- **Relocation:** only $\sim$0.5% of foreign-located inventors later appear in the US
- **Attrition:** no differential US inventor exit or entry

From Reduced Form to a Parameter: Why a Model?

What we found (Poisson DiD on inventors, no controls):

- International-only: $\approx -22\%$ Collaborative: $\approx -14\%$ Domestic-only: $\approx -7\%$ (n.s.)
- Pattern is consistent with domestic–foreign complementarity

From Reduced Form to a Parameter: Why a Model?

What we found (Poisson DiD on inventors, no controls):

- International-only: $\approx -22\%$ Collaborative: $\approx -14\%$ Domestic-only: $\approx -7\%$ (n.s.)
- Pattern is consistent with domestic–foreign complementarity

The model's job:

1. Which elasticity of substitution σ between domestic and foreign inventive effort rationalizes the missing domestic offset?
2. How large would σ have to be for the policy to **onshore** innovation?

From Reduced Form to a Parameter: Why a Model?

What we found (Poisson DiD on inventors, no controls):

- International-only: $\approx -22\%$ Collaborative: $\approx -14\%$ Domestic-only: $\approx -7\%$ (n.s.)
- Pattern is consistent with domestic–foreign complementarity

The model's job:

1. Which elasticity of substitution σ between domestic and foreign inventive effort rationalizes the missing domestic offset?
2. How large would σ have to be for the policy to **onshore** innovation?

Scope

A stylized patent-production model used as a *disciplining device*: a calibrated map from reduced-form moments to σ, not a full structural estimation

Model: Patent Production with Domestic and Foreign Inventors

Firm $i \in \{T, C\}$ with pre-shock effort $(\bar{e}_d, \bar{e}_f)$ chooses (e_d, e_f):

Model: Patent Production with Domestic and Foreign Inventors

Firm $i \in \{T, C\}$ with pre-shock effort $(\bar{e}_d, \bar{e}_f)$ chooses (e_d, e_f):

$$\max_{e_d, e_f} A_i \left[\alpha_i (e_d - c_d)^\rho + (1 - \alpha_i)(e_f - c_f)^\rho\right]^{\gamma/\rho} - (1 + \tau_d) e_d - (1 + \tau_f) e_f - \frac{\lambda}{2\bar{e}_d}(e_d - \bar{e}_d)^2 - \frac{\lambda}{2\bar{e}_f}(e_f - \bar{e}_f)^2$$

Model: Patent Production with Domestic and Foreign Inventors

Firm $i \in \{T, C\}$ with pre-shock effort $(\bar{e}_d, \bar{e}_f)$ chooses (e_d, e_f):

$$\max_{e_d, e_f} A_i\left[\alpha_i(e_d - c_d)^\rho + (1-\alpha_i)(e_f - c_f)^\rho\right]^{\gamma/\rho} - (1+\tau_d)e_d - (1+\tau_f)e_f - \frac{\lambda}{2\bar{e}_d}(e_d - \bar{e}_d)^2 - \frac{\lambda}{2\bar{e}_f}(e_f - \bar{e}_f)^2$$

Production

CES in *discretionary* effort $e_j - c_j$ (Stone–Geary minimum thresholds)

Without c_j, a homothetic CES gives *zero* DiD by construction - both types of inventors respond proportionally

With c_j, treated firms (large foreign effort) have more discretionary margin to re-optimize than control firms

$\sigma = 1/(1-\rho)$: substitution between domestic and foreign effort

γ: returns to scale

Model: Patent Production with Domestic and Foreign Inventors

Firm $i \in \{T, C\}$ with pre-shock effort $(\bar{e}_d, \bar{e}_f)$ chooses (e_d, e_f):

$$\max_{e_d, e_f} A_i\left[\alpha_i(e_d - c_d)^\rho + (1-\alpha_i)(e_f - c_f)^\rho\right]^{\gamma/\rho} - (1+\tau_d)e_d - (1+\tau_f)e_f - \frac{\lambda}{2\bar{e}_d}(e_d - \bar{e}_d)^2 - \frac{\lambda}{2\bar{e}_f}(e_f - \bar{e}_f)^2$$

Tax wedge

Effort proxied by R&D spend; §174 amortization raises per-unit cost

$\tau_d = 0.055$, $\tau_f = 0.11$ (exact: 5.4%, 12.4%)

Model: Patent Production with Domestic and Foreign Inventors

Firm $i \in \{T, C\}$ with pre-shock effort $(\bar{e}_d, \bar{e}_f)$ chooses (e_d, e_f):

$$\max_{e_d, e_f} A_i \left[\alpha_i (e_d - c_d)^\rho + (1-\alpha_i)(e_f - c_f)^\rho\right]^{\gamma/\rho} - (1+\tau_d)e_d - (1+\tau_f)e_f - \frac{\lambda}{2\bar{e}_d}(e_d - \bar{e}_d)^2 - \frac{\lambda}{2\bar{e}_f}(e_f - \bar{e}_f)^2$$

Adjustment costs

Proportional quadratic (Hayashi-style), scaled by $1/\bar{e}_j$

λ: curvature

Model: Patent Production with Domestic and Foreign Inventors

Firm $i \in \{T, C\}$ with pre-shock effort $(\bar{e}_d, \bar{e}_f)$ chooses (e_d, e_f):

$$\max_{e_d, e_f} A_i \left[\alpha_i (e_d - c_d)^\rho + (1-\alpha_i)(e_f - c_f)^\rho\right]^{\gamma/\rho} - (1+\tau_d)e_d - (1+\tau_f)e_f - \frac{\lambda}{2\bar{e}_d}(e_d - \bar{e}_d)^2 - \frac{\lambda}{2\bar{e}_f}(e_f - \bar{e}_f)^2$$

Mapping the model to the DiD estimates

Patents $\propto$ effort $\Rightarrow$ the model's change in $\log e_j$ is its predicted change in j-type patents (log points $\times 100$):

$$\beta_j = \left(\log e_j^T - \log \bar{e}_j^T\right) - \left(\log e_j^C - \log \bar{e}_j^C\right), \quad j \in \{d, f\}$$

Model: Patent Production with Domestic and Foreign Inventors

Firm $i \in \{T, C\}$ with pre-shock effort $(\bar{e}_d, \bar{e}_f)$ chooses (e_d, e_f):

$$\max_{e_d, e_f} A_i\left[\alpha_i(e_d - c_d)^\rho + (1-\alpha_i)(e_f - c_f)^\rho\right]^{\gamma/\rho} - (1+\tau_d)e_d - (1+\tau_f)e_f - \frac{\lambda}{2\bar{e}_d}(e_d - \bar{e}_d)^2 - \frac{\lambda}{2\bar{e}_f}(e_f - \bar{e}_f)^2$$

Mapping the model to the DiD estimates

Patents $\propto$ effort $\Rightarrow$ the model's change in $\log e_j$ is its predicted change in j-type patents (log points $\times 100$):

$$\beta_j = \left(\log e_j^T - \log \bar{e}_j^T\right) - \left(\log e_j^C - \log \bar{e}_j^C\right), \quad j \in \{d, f\}$$

Model (depends on σ)

$$\beta_d(\sigma), \quad \beta_f(\sigma)$$

$\Longleftrightarrow$

Data (patent-application Poisson DiD)

$$\hat{\beta}_d = -5.5, \quad \hat{\beta}_f = -18.7$$

Calibration: Matching the Domestic and Foreign Moments

Pre-shock effort (R&D/assets $\times$ pre-TCJA inventor share)

	Domestic $\bar{e}_d$	Foreign $\bar{e}_f$
Treated	4.0	5.0
Control	9.0	0.4

- Thresholds $c_d = c_f = 0.3$ (10th pctile of pre-shock effort)
- Treatment and control start with different allocations; observed mix treated as an interior optimum

Calibration: Matching the Domestic and Foreign Moments

Pre-shock effort (R&D/assets × pre-TCJA inventor share)

	Domestic $\bar{e}_d$	Foreign $\bar{e}_f$
Treated	4.0	5.0
Control	9.0	0.4

- Thresholds $c_d = c_f = 0.3$ (10th pctile of pre-shock effort)
- Treatment and control start with different allocations; observed mix treated as an interior optimum

Estimation

- Target: $\hat{\beta}_f = -18.7$, $\hat{\beta}_d = -5.5$ (patent-application Poisson DiD, no controls)
- Grid over (γ, λ); for each, choose $\sigma \in (0, 7]$ minimizing distance to the empirical pair
- Best fit: $\gamma = 0.76$, $\lambda = 0.16$, $\hat{\sigma} = 0.552$
- Predicted: $\beta_d = -5.49$, $\beta_f = -18.7$

Calibration: Matching the Domestic and Foreign Moments

Pre-shock effort (R&D/assets × pre-TCJA inventor share)

	Domestic $\bar{e}_d$	Foreign $\bar{e}_f$
Treated	4.0	5.0
Control	9.0	0.4

- Thresholds $c_d = c_f = 0.3$ (10th pctile of pre-shock effort)
- Treatment and control start with different allocations; observed mix treated as an interior optimum

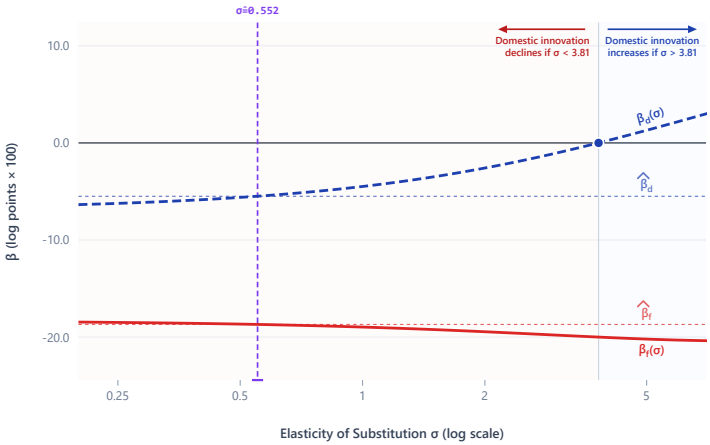
Estimation

- Target: $\hat{\beta}_f = -18.7$, $\hat{\beta}_d = -5.5$ (patent-application Poisson DiD, no controls)
- Grid over (γ, λ); for each, choose $\sigma \in (0, 7]$ minimizing distance to the empirical pair
- Best fit: $\gamma = 0.76$, $\lambda = 0.16$, $\hat{\sigma} = 0.552$
- Predicted: $\beta_d = -5.49$, $\beta_f = -18.7$

Result

$\hat{\sigma} = 0.552$: domestic and foreign inventive effort are **complements**

Onshoring Requires $\sigma > 3.8$; the Data Say $\hat{\sigma} \approx 0.55$



- Simulated DiD vs. σ, other parameters fixed
- Foreign effect ≈ -19 to -20 log points for any σ
- Domestic effect turns positive only above $\sigma = 3.81$
- $\hat{\sigma} = 0.55 \ll 3.8$: policy design could not onshore
- Every (γ, λ) with fit error < 0.2 gives $\sigma < 0.85$

OBBBA (2025) entrenches and increases the wedge

- Restored immediate full expensing for domestic R&D (tax years after Dec. 31, 2024)
- Foreign R&D: still 15-year amortization
- $\rightarrow$ *increases* wedge between domestic and foreign R&D
- Continued drag on domestic innovation via complementarities

OBBBA (2025) entrenches and increases the wedge

- Restored immediate full expensing for domestic R&D (tax years after Dec. 31, 2024)
- Foreign R&D: still 15-year amortization
- $\rightarrow$ *increases* wedge between domestic and foreign R&D
- Continued drag on domestic innovation via complementarities
- But permanence may (finally) make increased domestic R&D investments optimal

OBBBA (2025) entrenches and increases the wedge

- Restored immediate full expensing for domestic R&D (tax years after Dec. 31, 2024)
- Foreign R&D: still 15-year amortization
- $\rightarrow$ *increases* wedge between domestic and foreign R&D
- Continued drag on domestic innovation via complementarities
- But permanence may (finally) make increased domestic R&D investments optimal
- The answer to that will take time...

Contributions

- **1/3: Location of Inventive Activity**
  How the *geography* of innovation (not just its level) responds to a tax-induced user-cost wedge, using inventor-level patent data separated into domestic-only, collaborative, and international teams.

  Location of R&D: Wilson (2009); Moretti & Wilson (2017); Akcigit, Baslandze & Stantcheva (2016); Knoll et al. (2021); Huang, Krull & Ziedonis (2020) · §174 levels: Cowx et al. (2025); Du & Li (2025)

Contributions

- **2/3: Domestic vs. Foreign R&D Substitutability**
  The domestic null disciplines the elasticity of substitution: $\hat{\sigma} \approx 0.55$, far below the $\sigma > 3.8$ onshoring requires. Mirror image of Glennon (2024), where raising the cost of the domestic input shifted work abroad.

  Capital and labor: Desai, Foley & Hines (2009); Kovak, Oldenski & Sly (2021); Chodorow-Reich et al. (2024); Harrison & McMillan (2011) · Inventive inputs: Bilir & Morales (2020); Abramovsky, Griffith & Miller (2017)

Contributions

- **3/3: Unintended Consequences for Cross-Border Teams**
  When innovation is organized through cross-border teams, taxing the foreign input contracts joint knowledge production without a domestic offset.

  Kerr & Kerr (2018); Jaravel, Petkova & Bell (2018); Jones (2009); Foley & Kerr (2013); Huang, Krull & Ziedonis (2020)

Conclusions: The Cost of Onshoring R&D

The Policy Goal: TCJA §174 aimed to onshore R&D by creating a user-cost wedge between domestic (5-year) and foreign (15-year) amortization.

Conclusions: The Cost of Onshoring R&D

The Policy Goal: TCJA §174 aimed to onshore R&D by creating a user-cost wedge between domestic (5-year) and foreign (15-year) amortization.

The Result: The policy **failed to onshore innovation**.

- Treated firms cut international inventor activity by ≈18–22%; collaborative teams by ≈12–14%
- Domestic-only activity: no statistically significant change (-3% to -7%), no offset

Conclusions: The Cost of Onshoring R&D

The Policy Goal: TCJA §174 aimed to onshore R&D by creating a user-cost wedge between domestic (5-year) and foreign (15-year) amortization.

The Result: The policy **failed to onshore innovation**.

- Treated firms cut international inventor activity by ≈18–22%; collaborative teams by ≈12–14%
- Domestic-only activity: no statistically significant change (-3% to -7%), no offset

The Model: Calibration implies $\hat{\sigma} \approx 0.55$ (complements); onshoring would require $\sigma > 3.8$.

Conclusions: The Cost of Onshoring R&D

The Policy Goal: TCJA §174 aimed to onshore R&D by creating a user-cost wedge between domestic (5-year) and foreign (15-year) amortization.

The Result: The policy **failed to onshore innovation**.

- Treated firms cut international inventor activity by ≈18–22%; collaborative teams by ≈12–14%
- Domestic-only activity: no statistically significant change (-3% to -7%), no offset

The Model: Calibration implies $\hat{\sigma} \approx 0.55$ (complements); onshoring would require $\sigma > 3.8$.

The Bottom Line

Using the tax code to penalize foreign R&D does not automatically bring innovation home. When domestic and foreign inventors are complements, it contracts cross-border knowledge production without a domestic offset.

Conclusions: The Cost of Onshoring R&D

The Policy Goal: TCJA §174 aimed to onshore R&D by creating a user-cost wedge between domestic (5-year) and foreign (15-year) amortization.

The Result: The policy **failed to onshore innovation**.

- Treated firms cut international inventor activity by ≈18–22%; collaborative teams by ≈12–14%
- Domestic-only activity: no statistically significant change (-3% to -7%), no offset

The Model: Calibration implies $\hat{\sigma} \approx 0.55$ (complements); onshoring would require $\sigma > 3.8$.

The Bottom Line

Using the tax code to penalize foreign R&D does not automatically bring innovation home. When domestic and foreign inventors are complements, it contracts cross-border knowledge production without a domestic offset.

Thank you!
Questions?

`https://bowen.finance`